

Steps for Shifting to PQNK Farming on Flat Lands

NOTE

The below provided information should be read in conjunction with the sources listed at the end of this document. In case of any difference, the information provided in the sources supersedes this document.

1. **Make a Layout of PQNK Farm for the available land with following in mind:**
 - 1.1.1. Plan raised beds in **South-North** or North-South direction.
 - 1.1.2. Levelling of the land is required **only** in the direction of raised beds, so that the water flows under gravity in the water channels between the raised beds. In other directions no levelling is required.
 - 1.1.3. Water source should be on one side only.
 - 1.1.4. Plan for long fields as possible. Fields can be as long as 10 acres (2000 feet or 610 metres in length). Reduce/limit the length of the field to avoid excessive levelling.
 - 1.1.5. Keep entry and exit pathways for tractors and other machines.

1.1.6. Plan for drainage of excessive water in case of heavy rains or floods.

2. Survey the land for its topography to determine the minimum levelling requirements for the fields.

2.1.1. During the survey take readings at 04 different points in each acre.

3. Prepare the Land

3.1. Get your land tested to determine its PH level.

3.2. Remove any trees from the land, and demolish any existing watercourses/borders/boundaries around the fields.

3.3. Laser Level (if required) the land in the direction of planned raised beds to ensure that water flows under gravity. If required use Bulk Soil Shifter or Scraper.

3.4. Check the level of the field for good flow of water under gravity. Make a raised bed and its 02 water channels. Release water in the water channels and see that water flows smoothly without any hindrance. Otherwise, improve the levelling of the field. Also check if water seeps across the raised bed. If not, take actions to increase the water movement capacity of the soil.

3.5. Break Hardpan by deep ploughing with Chisel Plough or Subsoiler. Before deep

ploughing, dig in two or three points in the field to determine the depth and thickness of the hardpan. Generally it exists between 8 to 20 inches deep. Plough deeper than the hardpan with a distance of 5 feet between adjacent furrows/lines.

3.6. Prepare Long Fields so that tractor and other machinery conveniently operates in the fields. Long fields (up to 10 Acres) are preferable, but limit/reduce the length of the fields to avoid excessive levelling and/or soil disturbance, and efforts/resources available with you.

3.6.1. Make the water supply from one side only.

3.6.2. Make entry and exit pathways for tractors and other machines.

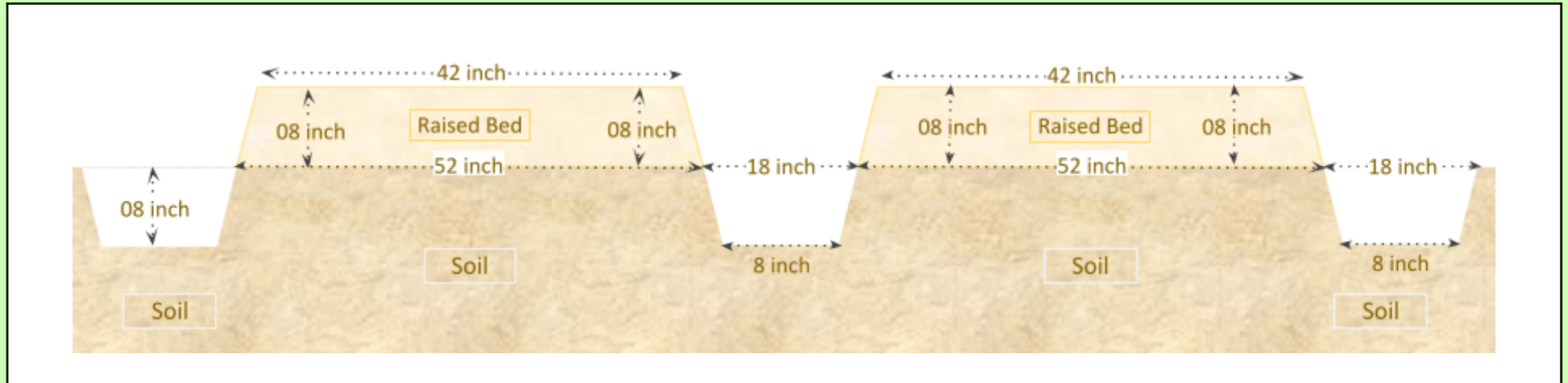
3.6.3. Make drainage for excessive rain/flood water. **How to make it?**

3.7. Heavily Water the land to leach down salts and lower the PH level. If the PH level of the land is 8 or above, mix 8-10 kilograms of Sulphuric Acid in the water.

3.8. Make Raised Beds

3.8.1. Wait for the appropriate moisture in the land. Plough the land well and make raised beds of the size as given in below picture with Bed Shaper or Rigger or any other local equipment. All beds must be straight and parallel with each

other. These are permanent beds. Do not plough or till these again. Only reshaping may be done, as and if required.



4. Place Organic Mulch on the raised beds to cover these completely.

- 4.1. Check that there is appropriate moisture in the land. If needed, release water in the newly made water channels. Fill the water channels 1/2 to 3/4 of their height.
- 4.2. On the newly-made raised beds grow a cover crop to improve the soil. These crops can be Jantar (جنتر), Lucerne (لوسن), Alfalfa, Peas (مٹر), Cowpea (گوارا), Radish (مولی), and other leguminous plants.

- 4.3. As the cover crop grows to some size ??, operate a Roller-Crimper on it to slash it as live-mulch to cover the raised beds. Do not uproot the plants, and let the roots remain in the soil to improve its water absorption capacity.
- 4.4. If cover crop is not possible, cover the new raised beds with any organic material or surplus agri-residue. This is dead-mulch. The layer of organic material should be 3-4 inch thick, and beds must be covered completely. Live-mulch is better than dead-mulch.

5. First and subsequent cash crops

- 5.1. As soon as the beds are covered with mulch, check for suitable moisture and plant the first cash crop with no-tilling planter. To check the moisture at any place, remove the mulch and dig about 2-3 inches, and observe the moisture level.
- 5.2. Multi-cropping (if feasible) can also be undertaken while growing cash crops.
- 5.3. As and when required, fill the water channels between the raised beds with water to their 1/2 to 3/4 height with water to increase the moisture in the soil.
- 5.4. Harvest the crop when ready, leave residue on the raised beds, plant the next crop without any ploughing or tilling.

Source:

1. <https://youtu.be/naklRXVA7cE?si=udv4jNnUYbfyOBWI>
2. <https://www.youtube.com/watch?v=XHwdjn6cG4k>
3. https://www.youtube.com/watch?v=nxSe_V9MP4c&t=2s